EarthCARE MAAP Hackathon
ESA MAAP & Data Access Introduction

31/03/2026

Saskia Brose

EarthCARE MAAP Hackathon 2026

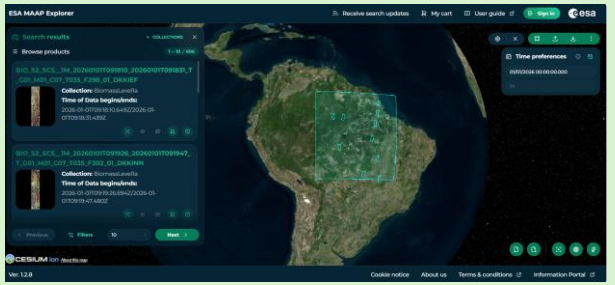
ESA MAAP Overview

ESA MAAP
Data Access

Catalogue

OPENAPI STAC OpenSearch

Explorer - GUI



ESA MAAP
Collaborative Environment

CODING

PROCESSING

Algos/ Processors

Scalable Resources

Other Services

VISUALISATION & ANALYTICS

ESA MAAP Overview

ESA MAAP Data Access

CATALOGUE

STAC

```
search = catalog.search(
    collections=EC_COLLECTION,
    filter="productType = 'MSI_COP_2A' and orbitNumber > 5000", # For examp
    bbox = [0, -20, 10, -10], # Filter by area
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items = list(search.items()) # Get all items as a list
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print(f"{results} items found that matched the query.")
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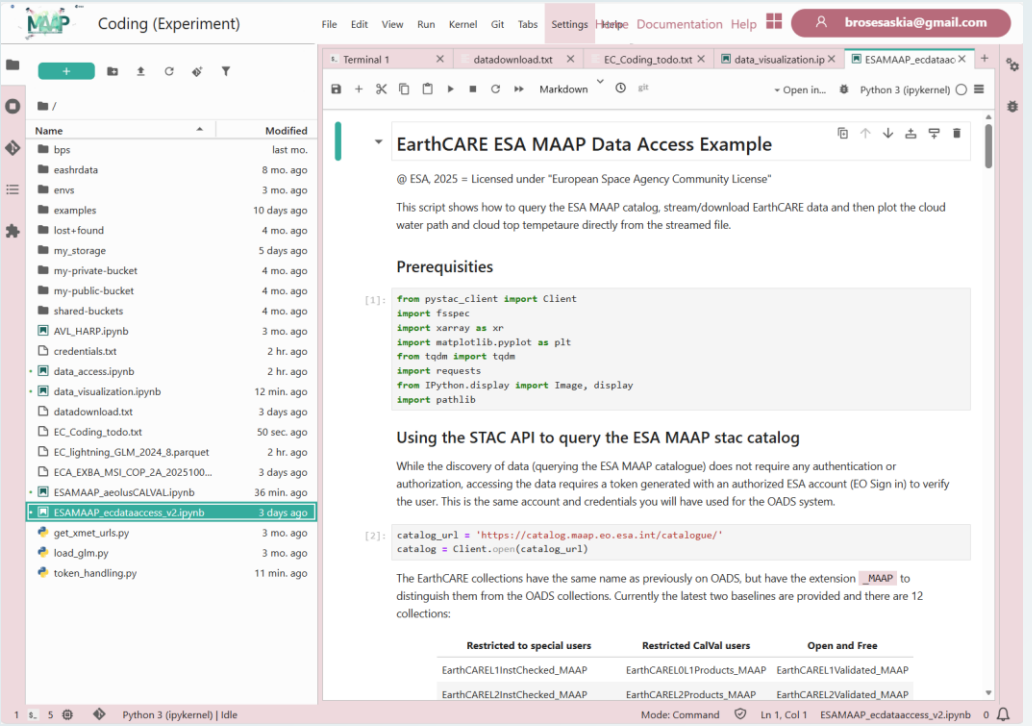
```

ESA MAAP Collaborative Environment

CODING



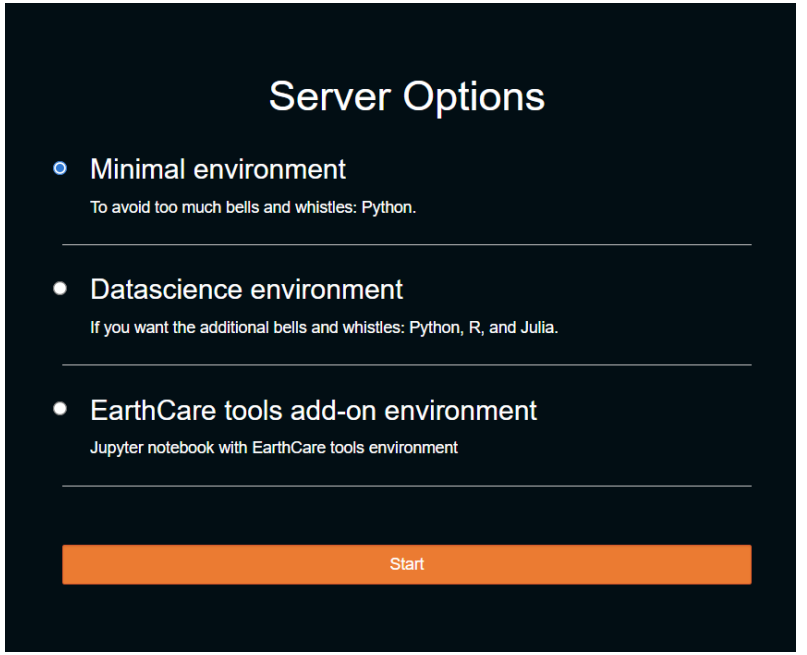
EarthCARE Collaborative Environment

https://portal.maap.eo.esa.int/earthcare/

- Join the Collaborative Environment: Join
- Accessing the JupyterLab: Tools > Product Algorithm Laboratory > Coding (Experiment)

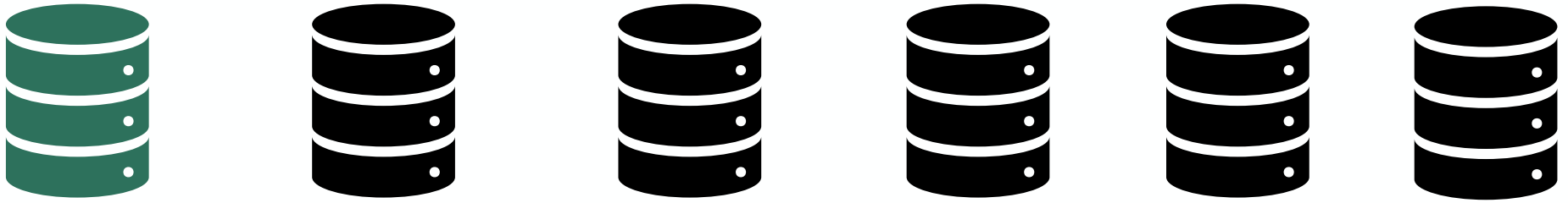
- Different server options in the JupyterLab:
  - Please select EarthCare tools add-on environment

- Mounted storages within the JupyterLab:
  - /home/jovyan
  - my-private-bucket
  - my-public-bucket <--> shared-buckets
  - my_storage
  - eashrdata



EarthCARE Collaborative Environment

https://experiment.earthcare.pal.maap.eo.esa.int/



/home/jovyan | my-private-bucket | my-public-bucket | eashrdata | my_storage | shared-buckets

Fast cloud storage

- ➔ CODE
- ➔ CONDA envs
- ➔ GIT

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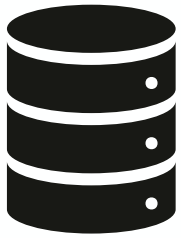


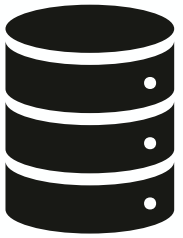
/home/jovyan

Fast cloud storage

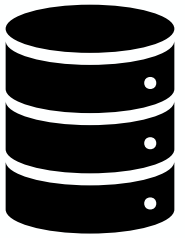
my-private-bucket

Large cloud storage

- ➔ Slow
- ➔ Storage space virtually unlimited
- ➔ Data files



my-public-bucket

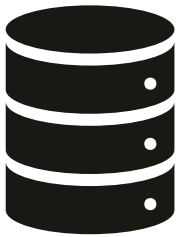


eashrdata

my_storage

Large cloud storage

- ➔ Connected to rest of PAL
- ➔ Processing outputs
- ➔ Uploaded data



shared-buckets

EarthCARE Collaborative Environment

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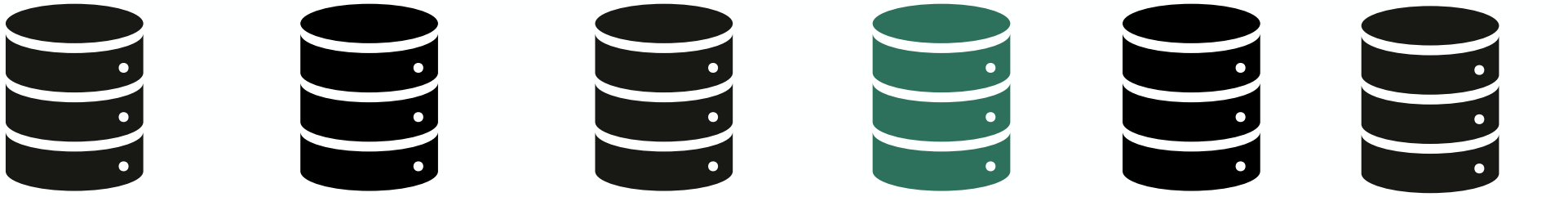

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Fast cloud storage

my-private-bucket

Large cloud storage

my-public-bucket

eashrdata

my_storage

Large cloud storage

shared-buckets

***Nextcloud**: Shared space for specific groups, fast upload of large files*

EarthCARE Collaborative Environment

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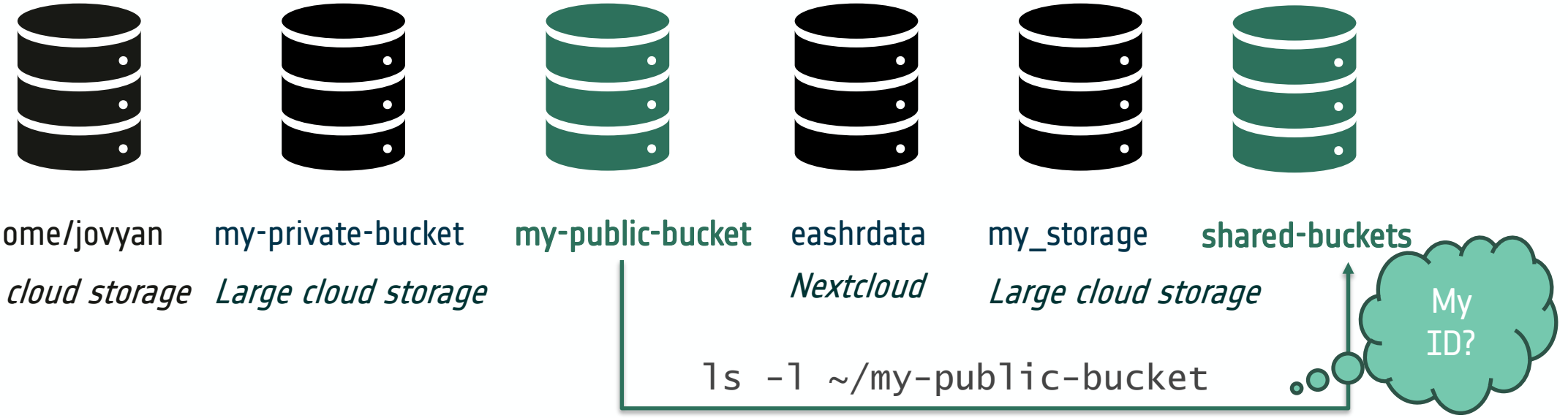

EarthCARE Data Access Example - Two step approach

Unpacked products hosted on ESA MAAP to enable

- Downloading single file within a product
- Reading/Streaming full products or subsets of products **without needing to download**

1) Select EarthCARE collection

2) Temporal-spatial search + Queryables to find individual data products

EarthCARE Data Access Example - Two step approach

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Fetch and use your Token
(user's authentication and authorization)

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ESA MAAP - offline token generation

eyJhbGciOiJIUzUxMiIsInR5cCIgOiAiSldUIiwia2lkIiA6ICJjOT
VmMGU4My02YjZkLTQ2YjEtOWJjZi1kZGUzZDdiOTY3N2
QifQ.eyJleHAiOjE3ODI3NDA5OTMsImlhdCI6MTc3NDg3O
DU5MywianRpIjoiMjYxYzU5YzEtNGVhZC00NTc0LTg2ZG
MtZTU1MzE1NjIwMjZmIiwiaXNzIjoiaHR0cHM6Ly9pYW0u

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    collecti
    filter="
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    max_item
)
items = list
results = se

print(f"{res
print(f"Acce
```

ESA MAAP - offline token generation

eyJhbGciOiJIUzUxMiIsInR5cCIgOiAiSldUIiwia2lkIiA
VmMGU4My02YjZkLTQ2YjEtOWJjZi1kZGUzZDdiOTY3N2
QifQ.eyJleHAiOjE3ODI3NDA5OTMsImlhdCI6MTc3NDg3O
DU5MywianRpIjoiMjYxYzU5YzEtNGVhZC00NTc0LTg2ZG
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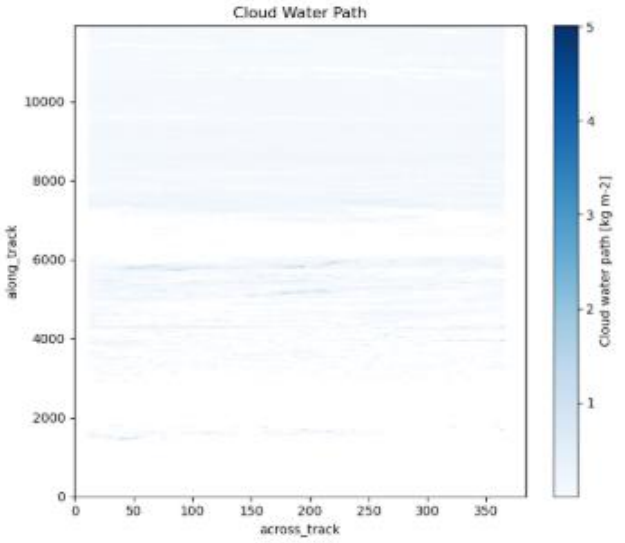
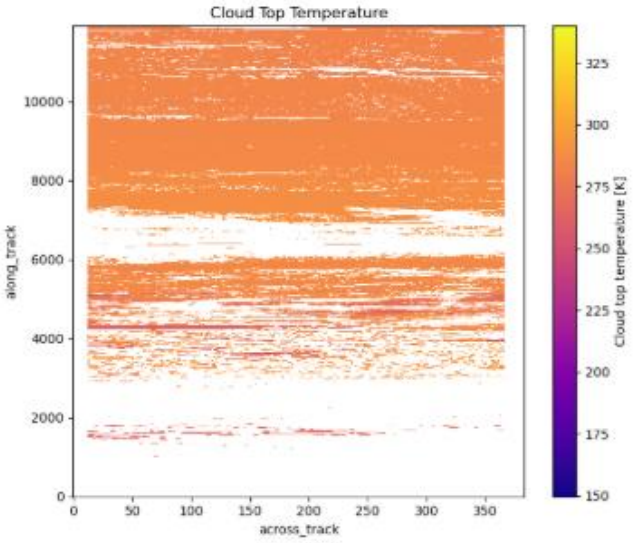
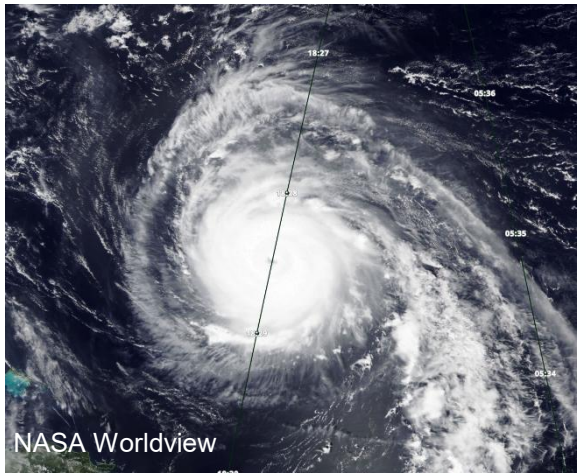
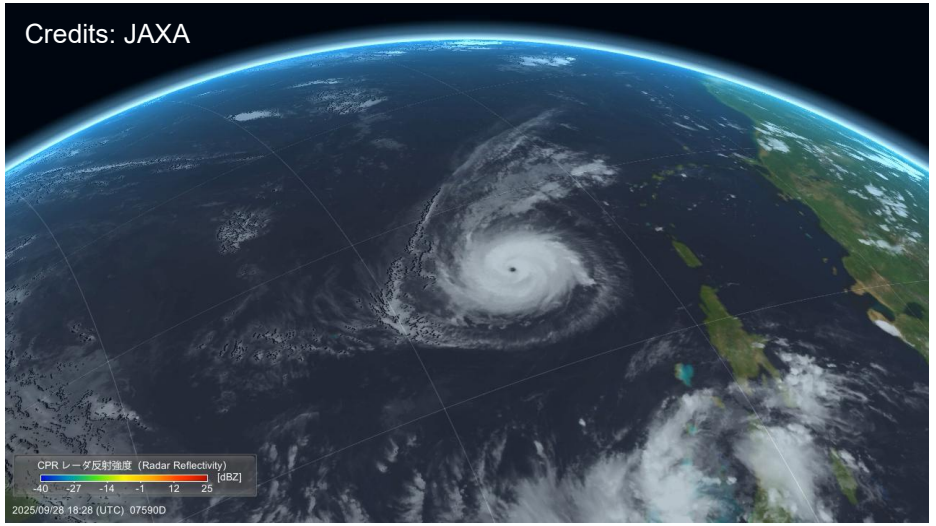


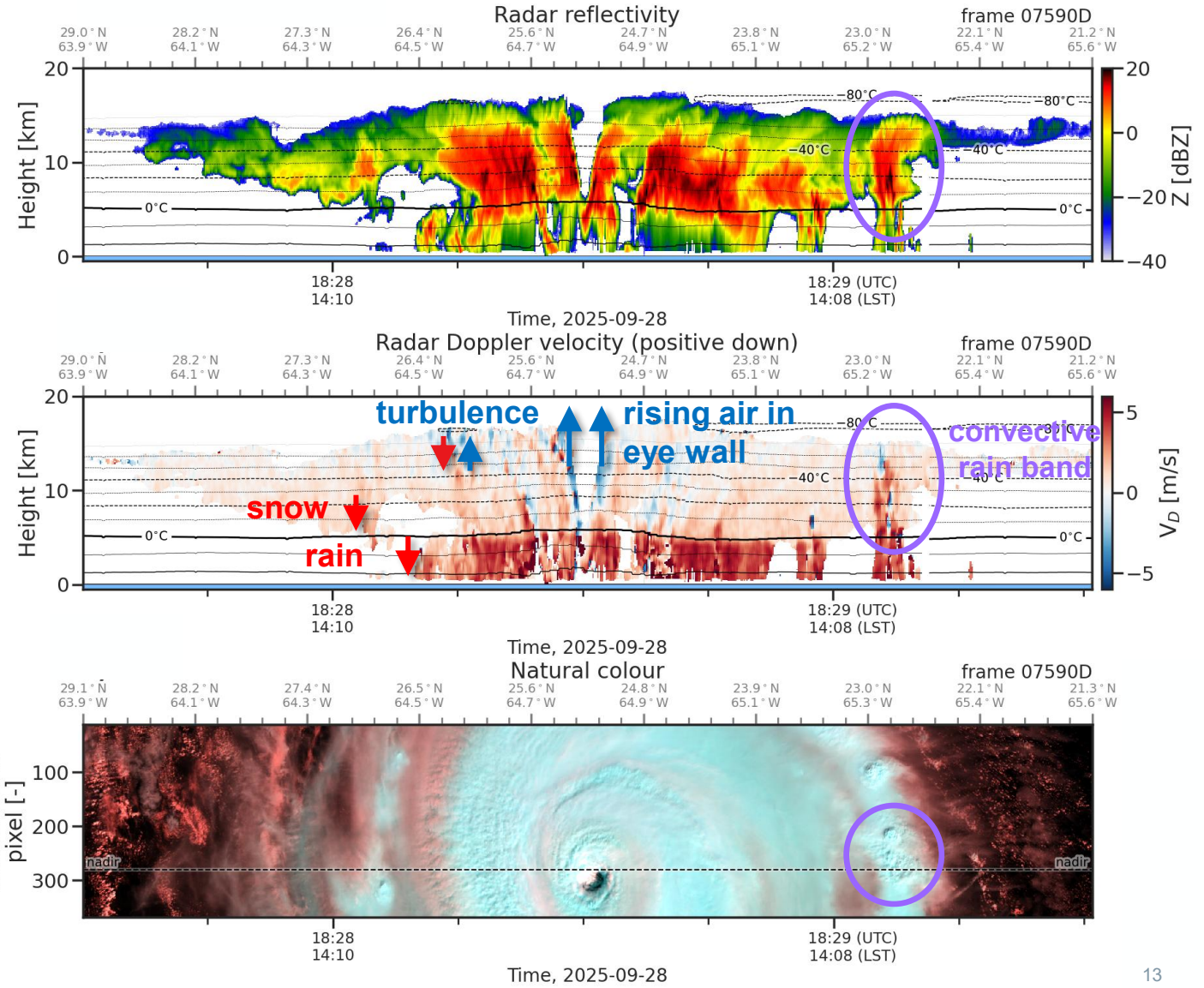


Read and Plot with fsspec & h5py
(no need to download)

EarthCARE Data Access example – hurricane Humberto







earthcarescience.net/gallery/tropical-cyclones

Where to find some data access examples:

My shared-bucket: 0c27e7de-7e0e-4f7d-8997-85e1092546a4

Data Access Example Scripts — ESA MAAP

Collaborative Environm... | ESA MAAP Knowledge-base

Thank you for your attention
Questions?